

NETWORKS AND COMPLEXITY

Solution 23-12

*This is an example solution from the forthcoming book *Networks and Complexity*.*

Find more exercises at <https://github.com/NC-Book/NCB>

Ex 23.12: Kirchhoff proof [Lvl. 5]

Find a forward-facing derivation of Kirchhoff's theorem and the matrix-tree theorem, i.e. a line of reasoning that leads to the result, rather than postulating the result and then proving that it is true.

Solution

This is an unsolved problem. Kirchhoff's own work only states the theorem without proof, but several subsequent papers have provided proofs, showing that the theorem is true. However, these proofs do not provide much intuition that would motivate the theorem.